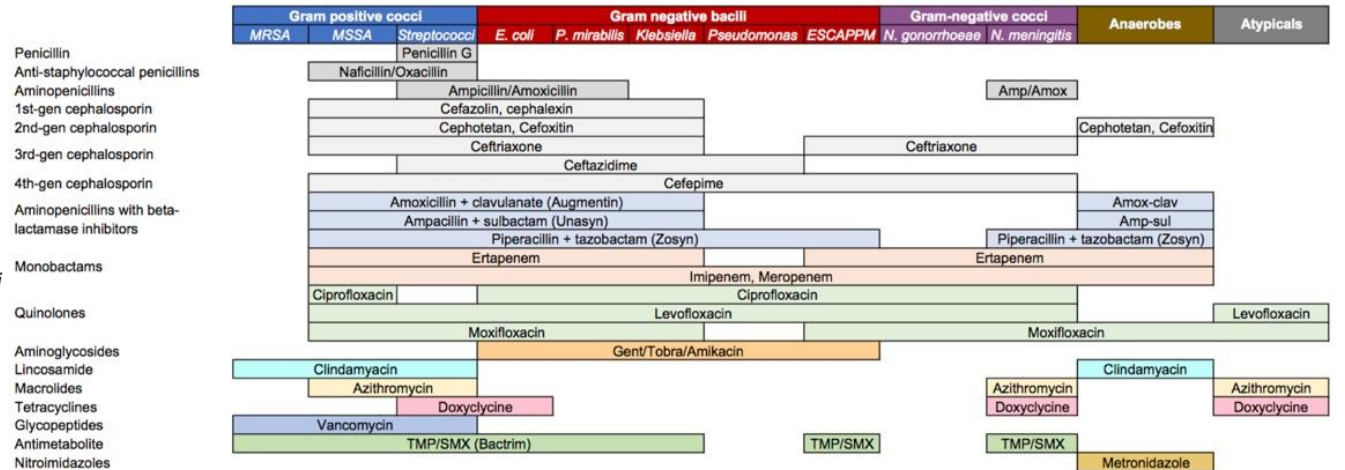
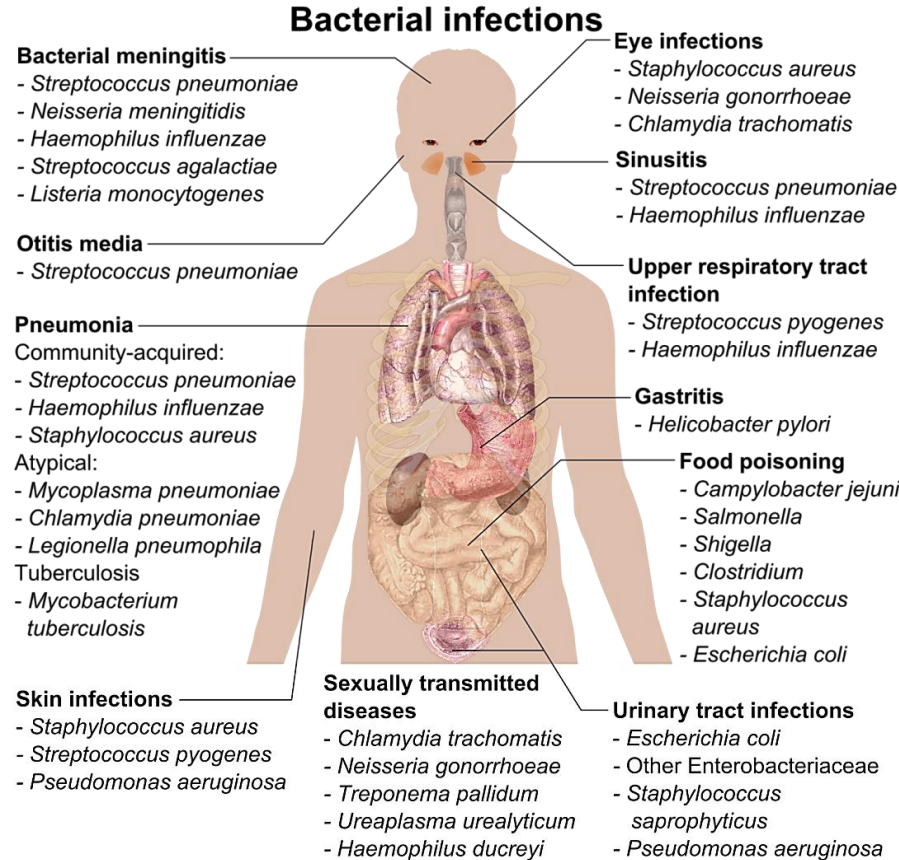


Antibacterial Therapy



See github.com/aetherist/antibiogram for details. For educational purposes only. TMP/SMX = Trimethoprim-sulfamethoxazole, MRSA = Methicillin-resistant *Staphylococcus aureus*, MSSA = Methicillin-sensitive *Staphylococcus aureus*, ESCAPPM = Enterobacter spp., Serratia spp., Citrobacter freundii, Aeromonas spp., Proteus spp., Providencia spp. and Morganella morganii.

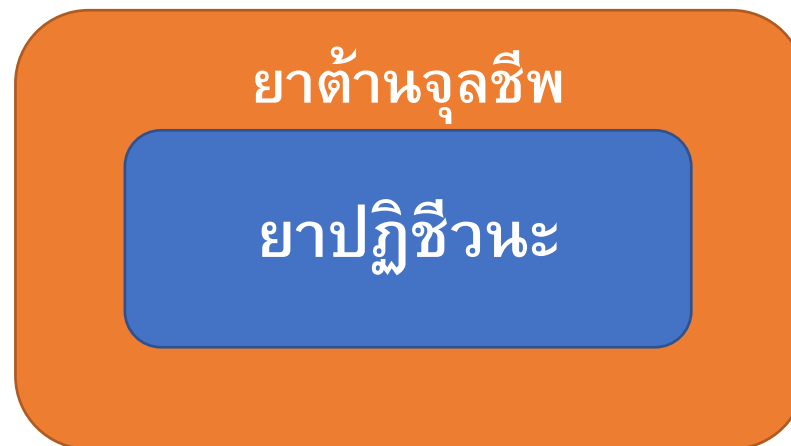
Arnatchai Maiuthed, Ph.D.

Department of Pharmacology, Faculty of Pharmacy, Mahidol University

Antimicrobial VS Antibiotic

ยาปฏิชีวนะ (antibiotic): สารที่สร้างขึ้นและแยกได้จากเชื้อจุลินทรีย์ชนิดหนึ่ง ที่สามารถออกฤทธิ์ยับยั้งการเจริญเติบโต หรือ ทำลาย เชื้อจุลินทรีย์อีกกลุ่มหนึ่งได้

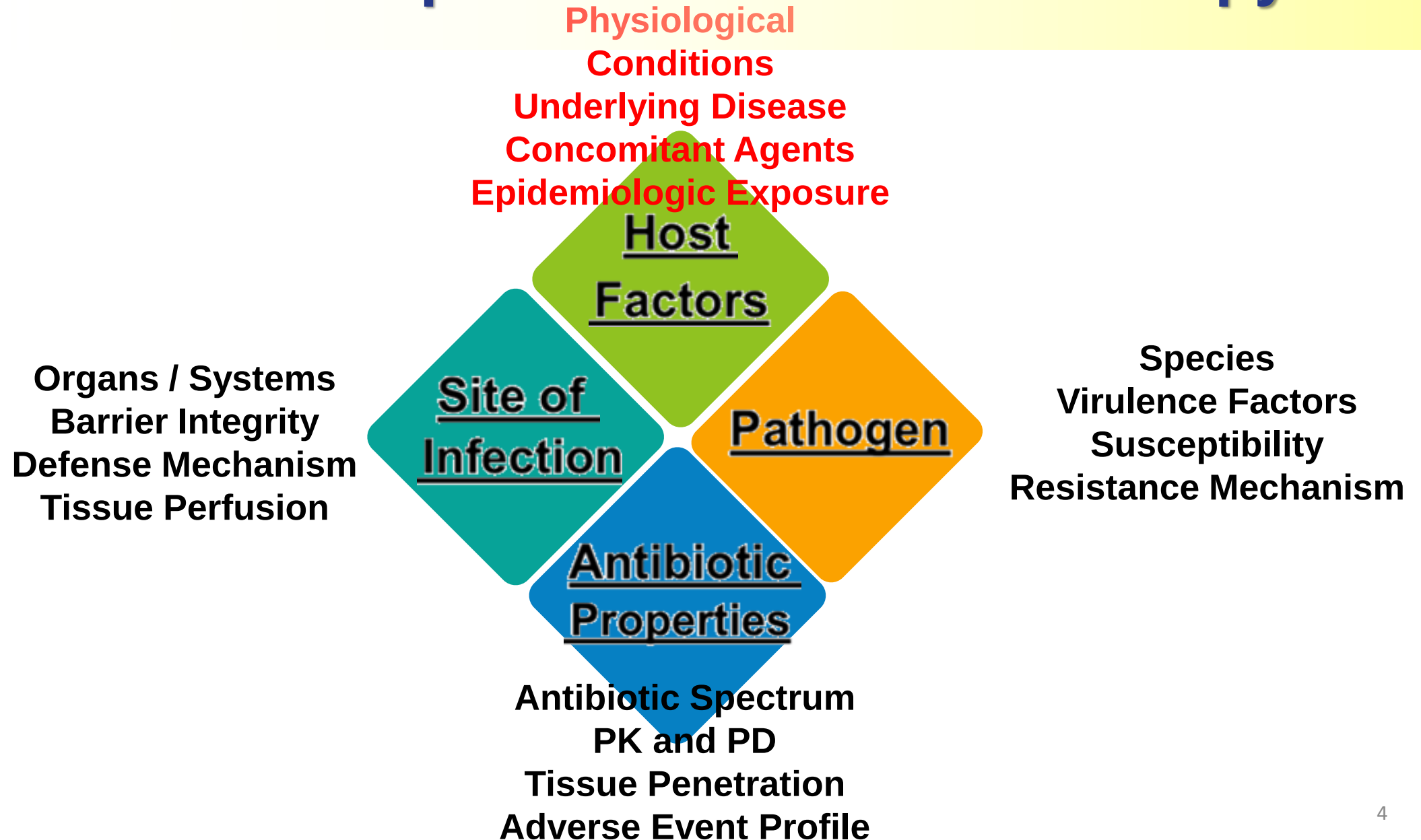
ยาต้านจุลชีพ (antimicrobial agents): สารที่แยกได้จากเชื้อจุลินทรีย์ **หรือ** สารกึ่งสังเคราะห์จากสารที่แยกได้จากเชื้อจุลินทรีย์ (ทั้งที่มีความเหมือนและต่างจากสารที่ได้จากเชื้อจุลินทรีย์) **หรือ** สารสังเคราะห์ทางเคมี ที่มีฤทธิ์ยับยั้งการเจริญเติบโต หรือ ทำลายเชื้อจุลินทรีย์



Goal of Antimicrobial Therapy

- Antimicrobial therapy should be **selectively toxicity**
 - Drugs should kill or inhibit microbial cells **without** simultaneously **damaging host tissues**
 - Targets **genetic** or **biochemical pathway** that unique to pathogen (e.g. bacterial cell wall synthesis inhibitor)
 - Target **protein isoform** that is unique to pathogen (e.g. dihydrofolate reductase inhibitor)

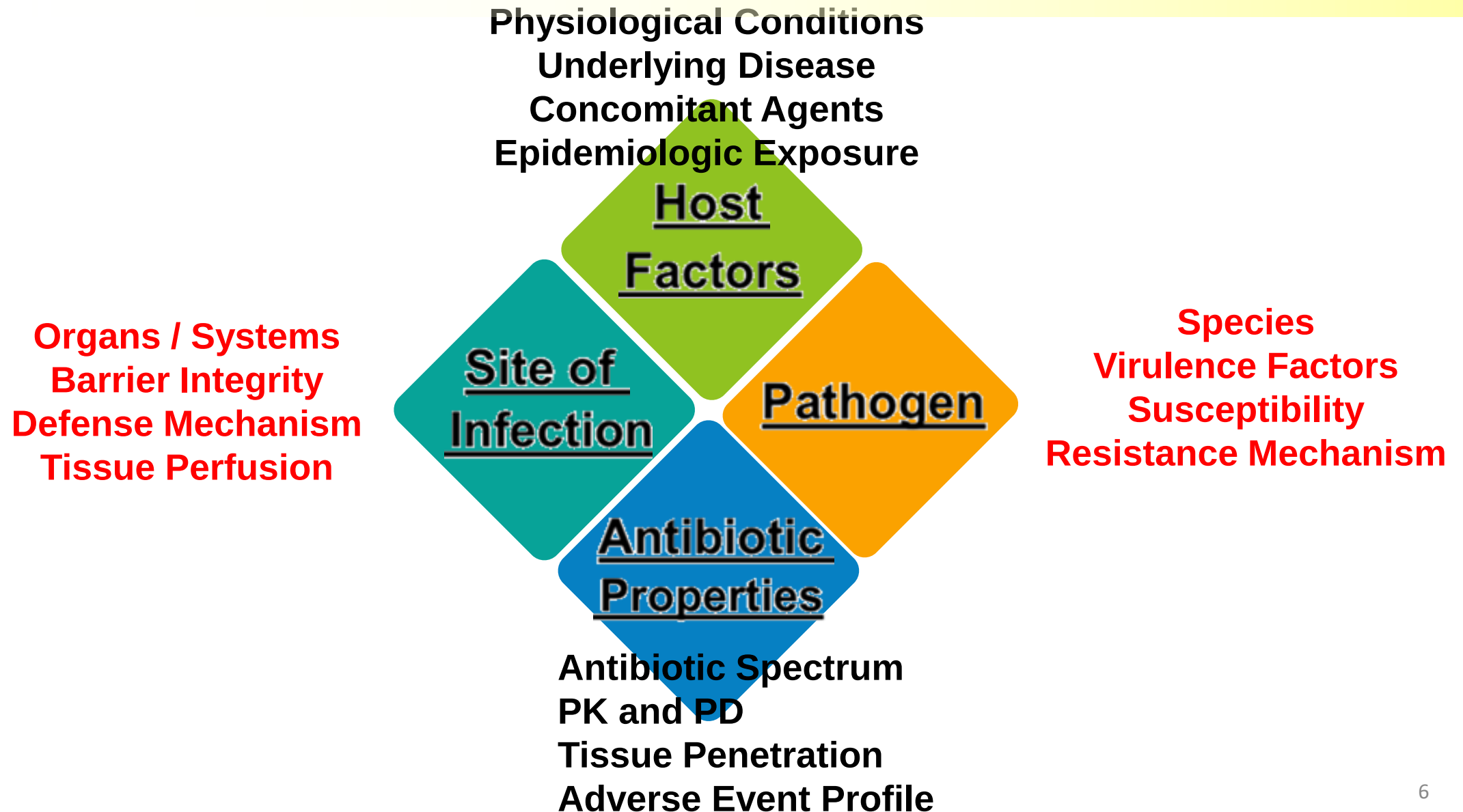
Principle of Antimicrobial Therapy



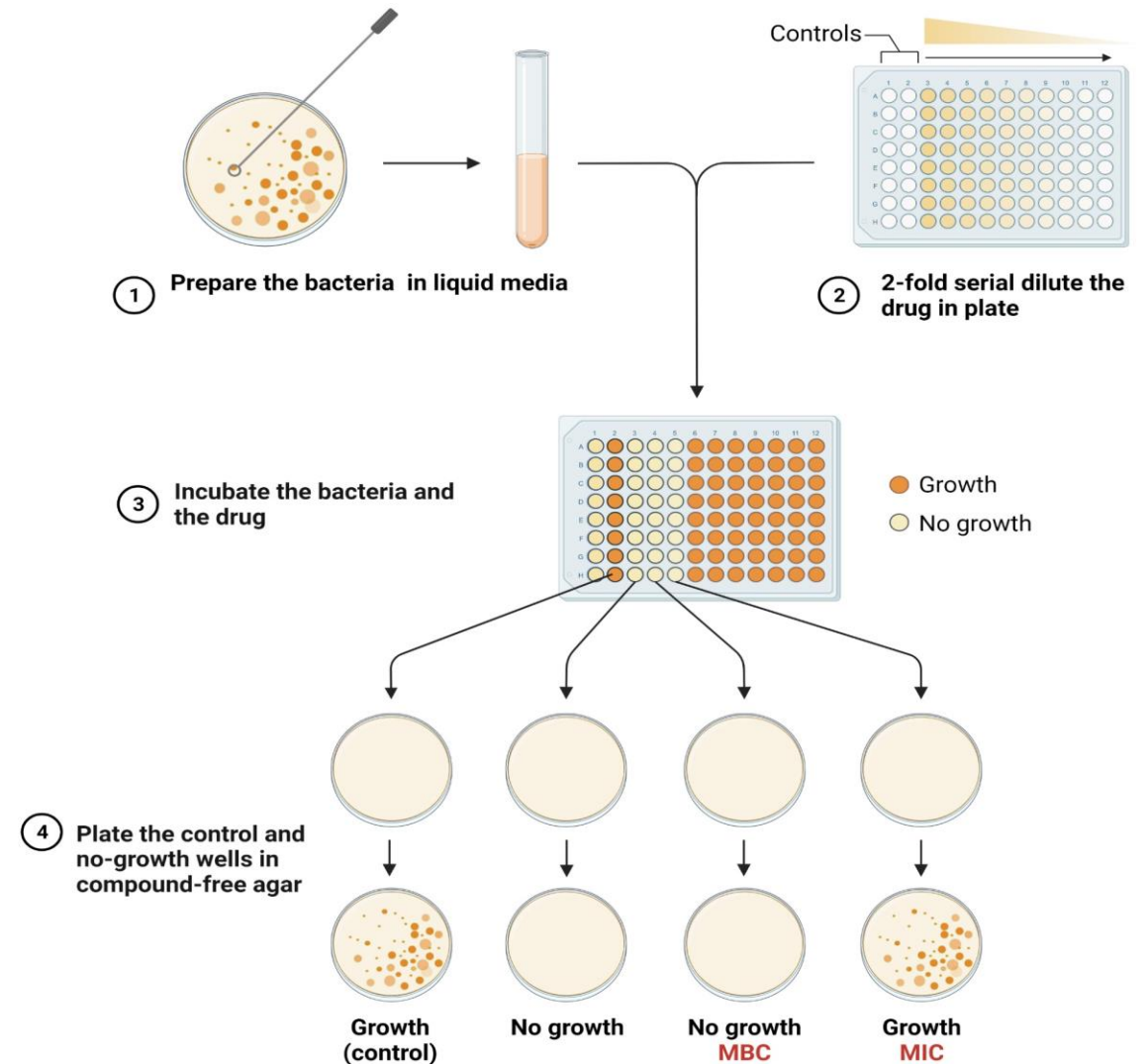
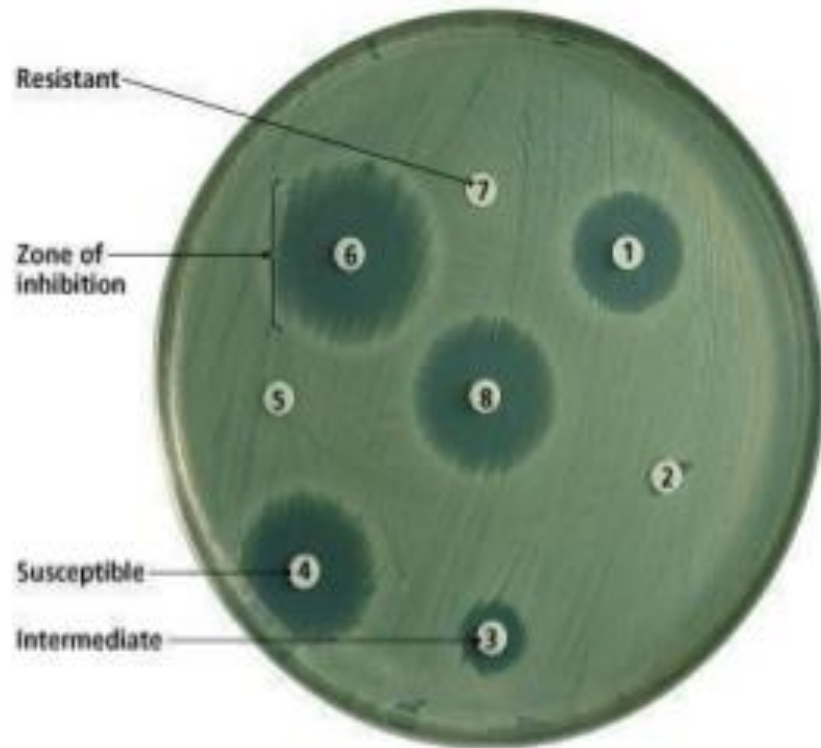
Host Factors

- **Physiological Conditions**
 - Newborn, Pediatric, Elderly, Pregnancy, Lactation
 - Hypersensitivity → can be severe or life threatening
- **Genetic**
 - G6PD deficiency
- **Underlying Disease**
 - AIDs, Cancer, Organ transplantation, Diabetes, Malnutrition, Certain genetic disorders → Immunocompromised
 - Kidney and Liver disease → Impaired elimination or Detoxification
- **Concomitant Agents**
 - Immunosuppressive medication
 - Drug interaction: CYP450 inhibitor/inducers, Antacid, Probenecid
 - Adverse effect profile: Synergistic AEs
- **Epidemiologic Exposure**
 - Sick family member, New sexual partner
 - Recent hospitalization, Recent Traveling
 - Previous antibiotic exposure → Trend to have resistant organism

Principle of Antimicrobial Therapy



Bacterial Susceptibility



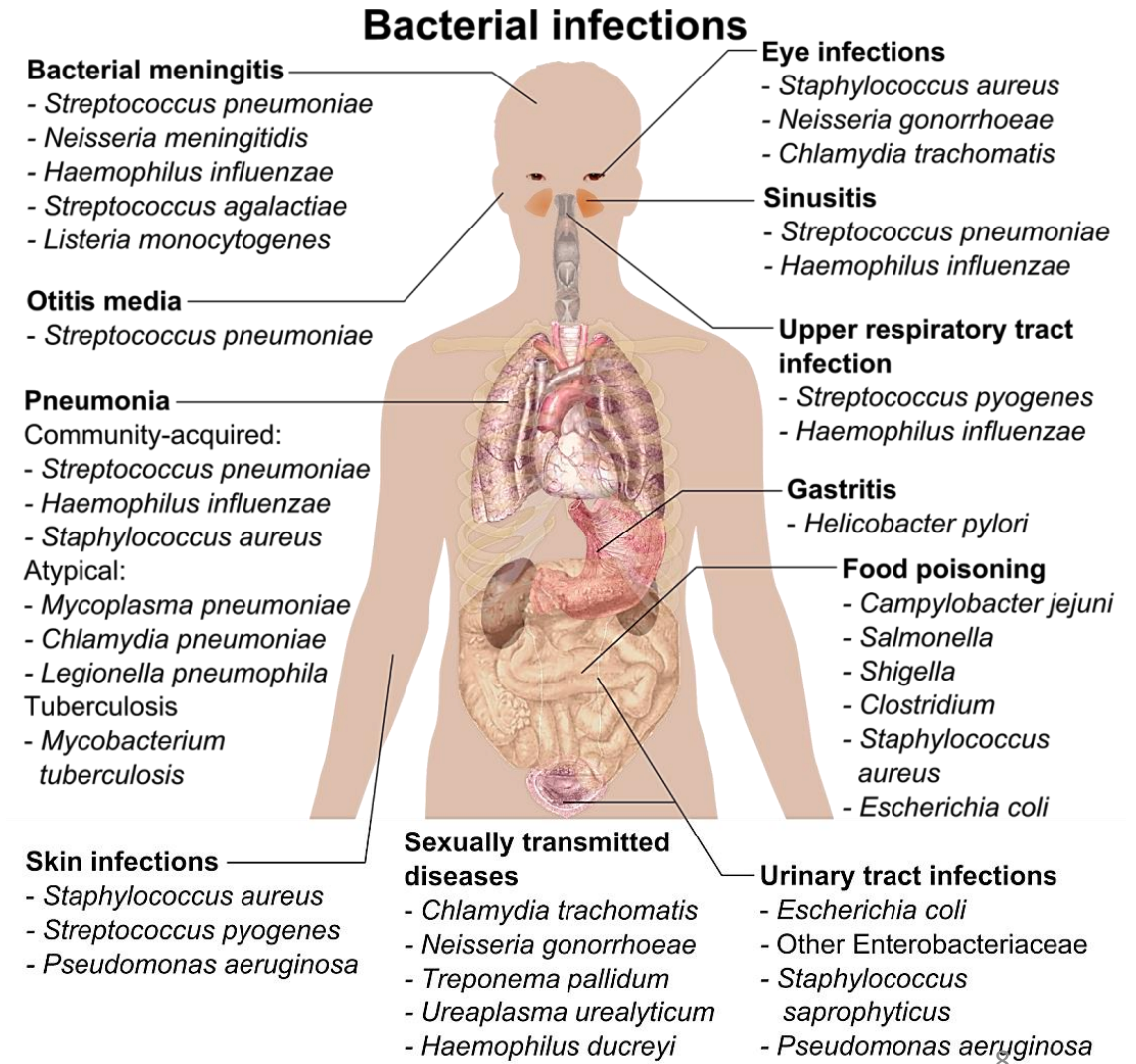
Results

MIC: Minimum inhibitory concentration. Bacteria cannot grow when incubated with the drug but can be rescued when plating in compound-free agar.

MBC: Minimum bactericidal concentration. Bacteria cannot grow when incubated with the drug and cannot be rescued when plating in compound-free agar.

Site of Infection

- Certain bacteria commonly cause **infection in particular body sites or fluids**
- Site of infection and likely **causative organism**
 - Helpful in **empirical antimicrobial selection** (the culture are available based on some preliminary information)
- Determine **dose & route of administration** of antimicrobial
 - Serum concentration vs. Tissue concentration
 - Relationship to MIC



Site of Infection

1. ติดเชื้อจุลินทรีย์ชนิดอะไร

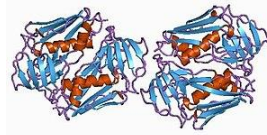
แบคทีเรีย

Gram positive:

Staphylococcus aureus. (microbiota-skin, URT),

Disease (ผิวหนัง (หนอง), ตา, ปอด, อาหารเป็นพิษ; **toxic shocked syndrome (TSS-fever**, rash, skin peeling, low BP), exotoxin (beta-hemolytic)

Methicillin-resistance *S. aureus* (MRSA):
worldwide problem



toxin shock syndrome toxin -1

https://en.wikipedia.org/wiki/Toxic_shock_syndrome

Streptococcus pyogenes. (**group A streptococcus**, **GAS**)

Disease: (ผิวหนัง, ทางเดินหายใจส่วนบน; **toxic-shock-like syndrome (TSLS)**, exotoxin (beta-hemolytic)- rash, fever, bio-film formation)

Overview of Bacterial infections

Bacterial meningitis

- *Streptococcus pneumoniae*
- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
- *Listeria monocytogenes*

Otitis media

- *Streptococcus pneumoniae*

Pneumonia

Community-acquired:

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Staphylococcus aureus*

Atypical:

- *Mycoplasma pneumoniae*
- *Chlamydia pneumoniae*
- *Legionella pneumophila*

Tuberculosis

- *Mycobacterium tuberculosis*

Skin infections

- *Staphylococcus aureus*
- *Streptococcus pyogenes*
- *Pseudomonas aeruginosa*

Eye infections

- *Staphylococcus aureus*
- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

- *Helicobacter pylori*

Food poisoning

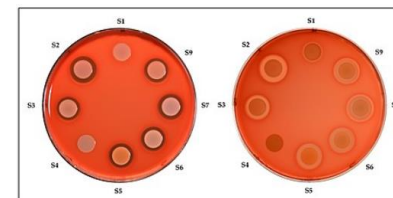
- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Sexually transmitted diseases

- *Chlamydia trachomatis*
- *Neisseria gonorrhoeae*
- *Treponema pallidum*
- *Ureaplasma urealyticum*
- *Haemophilus ducreyi*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*



Blood agar culture

https://en.wikipedia.org/wiki/File:Staphylococcus_aureus_identification.jpg

Site of Infection

1. ติดเชื้อจุลินทรีย์ชนิดอะไร

Gram positive:

Streptococcus pneumoniae. (pneumococcus),

Disease: sinusitis, pneumonia, hemolytic (alpha-aerobic, beta-anaerobic), facultative anaerobic,

Streptococcus agalactiae. (group B streptococcus, GBS), bacterial capsule (polysaccharide), microbiota-GI

Disease: (beta-hemolytic), neonatal infection

Overview of Bacterial infections

Bacterial meningitis

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- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
- *Listeria monocytogenes*

Otitis media

- *Streptococcus pneumoniae*

Pneumonia

Community-acquired:

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 - *Staphylococcus aureus*
- Atypical:

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 - *Legionella pneumophila*
- Tuberculosis
- *Mycobacterium tuberculosis*

Skin infections

- *Staphylococcus aureus*
- *Streptococcus pyogenes*
- *Pseudomonas aeruginosa*

Eye infections

- *Staphylococcus aureus*
- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Sexually transmitted diseases

- *Chlamydia trachomatis*
- *Neisseria gonorrhoeae*
- *Treponema pallidum*
- *Ureaplasma urealyticum*
- *Haemophilus ducreyi*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*

Site of Infection

1. ติดเชื้อจุลินทรีย์ชนิดอะไร
แบคทีเรีย

Gram negative:

Escherichia coli. (microbiota, large intestine: vitamin K, facultative anaerobe), Diseases; urinary tract infection,

Pseudomonas aeruginosa (encapsulated, facultative anaerobe); multidrug resistance pathogen (advance antibiotic resistance mechanisms), opportunistic infection (hospital acquired; cystic fibrosis, traumatic burn) and nosocomial infection, biofilm

Overview of Bacterial infections

Bacterial meningitis

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- *Listeria monocytogenes*

Otitis media

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Sexually transmitted diseases

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- *Ureaplasma urealyticum*
- *Haemophilus ducreyi*

Eye infections

- *Staphylococcus aureus*
- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*

Site of Infection

1. ติดเชื้อจุลชีพชนิดอะไร
แบคทีเรีย

Gram negative:

Haemophilus influenzae (facultative
anaerobe), opportunistic infection
(pneumonia, meningitis, URT infection)

Overview of Bacterial infections

Bacterial meningitis

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- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
- *Listeria monocytogenes*

Otitis media

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Pneumonia

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Eye infections

- *Staphylococcus aureus*
- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

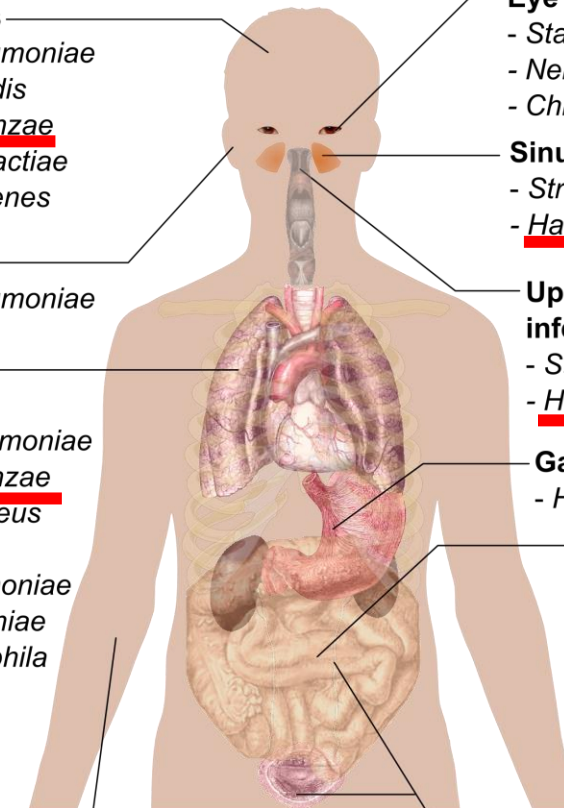
- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*



Site of Infection

1. ติดเชื้อจุลินทรีย์ชนิดอะไร แบคทีเรีย

Gram negative:

Neisseria gonorrhoeae.

Disease: STD, eye infection; survive
within neutrophile,

Campylobacter jejuni. (animal feces-bird),
helical shape with flagella

Disease: diarrhea to bloody-stools,
enterotoxin and cytotoxin,



<https://www.news-medical.net/health/What-is-Campylobacteriosis.aspx>

Overview of Bacterial infections

Bacterial meningitis

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- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
- *Listeria monocytogenes*

Otitis media

- *Streptococcus pneumoniae*

Pneumonia

Community-acquired:

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Staphylococcus aureus*

Atypical:

- *Mycoplasma pneumoniae*
- *Chlamydia pneumoniae*
- *Legionella pneumophila*

Tuberculosis

- *Mycobacterium tuberculosis*

Skin infections

- *Staphylococcus aureus*
- *Streptococcus pyogenes*
- *Pseudomonas aeruginosa*

Sexually transmitted diseases

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- *Neisseria gonorrhoeae*
- *Treponema pallidum*
- *Ureaplasma urealyticum*
- *Haemophilus ducreyi*

Eye infections

- *Staphylococcus aureus*
- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*

Site of Infection

1. ติดเชื้อจุลินทรีย์ชนิดอะไร
แบคทีเรีย

Gram negative:

Salmonella spp.

Shigella spp.

Clostridium spp.

Overview of Bacterial infections

Bacterial meningitis

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- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
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- *Neisseria gonorrhoeae*
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Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

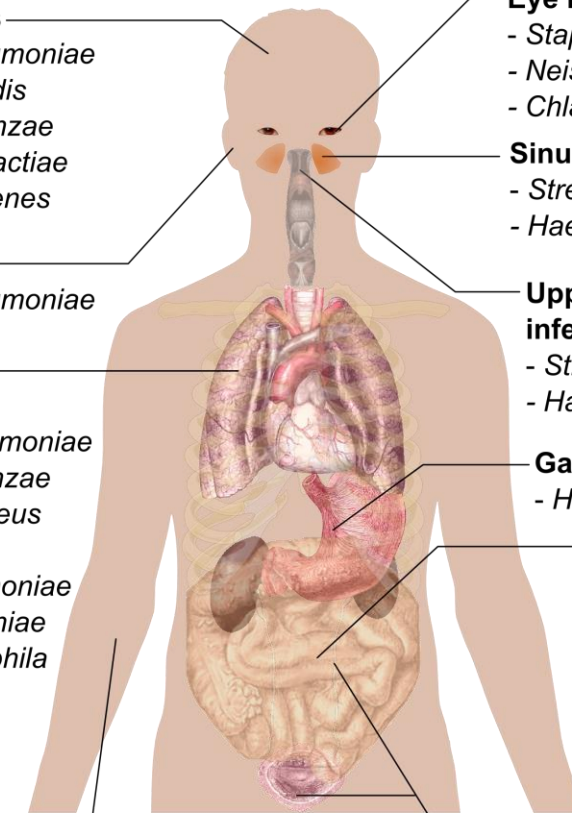
- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*



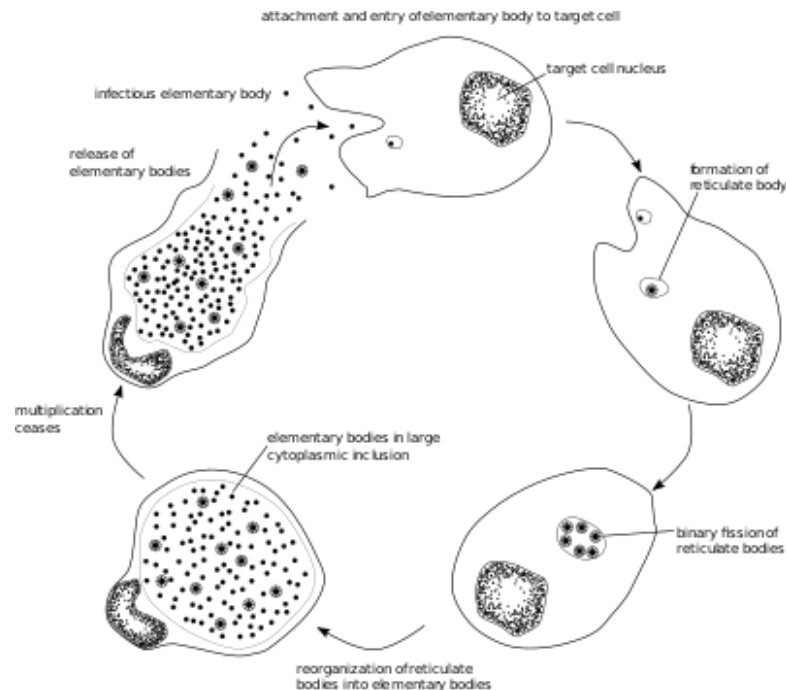
Site of Infection

1. ติดเชื้อจุลชีพชนิดอะไร
แบบที่เรื้อรัง

Intracellular bacteria:

Chlamydia trachomatis. Gram negative, (Elementary bodies-infect and survive outside host cell, Reticulate bodies-replicate only within host cell),

Disease: STD, eye infection (blindness, conjunctivitis)



https://en.wikipedia.org/wiki/Chlamydia_trachomatis

Overview of Bacterial infections

Bacterial meningitis

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Otitis media

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Pneumonia

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Tuberculosis

- *Mycobacterium tuberculosis*

Skin infections

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- *Pseudomonas aeruginosa*

Sexually transmitted diseases

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- *Chlamydia trachomatis*

Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*

Site of Infection

1. ติดเชื้อจุลชีพชนิดอะไร

แบคทีเรีย

Special bacteria:

Mycobacterium tuberculosis. Mycolic acid

(wax) (Gram +), require high oxygen, **slow growth rate**

Disease: intracellular pathogen, tuberculosis, macrophage infection

Overview of Bacterial infections

Bacterial meningitis

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- *Neisseria meningitidis*
- *Haemophilus influenzae*
- *Streptococcus agalactiae*
- *Listeria monocytogenes*

Otitis media

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Pneumonia

Community-acquired:

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Atypical:

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- *Legionella pneumophila*

Tuberculosis

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Skin infections

- *Staphylococcus aureus*
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Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

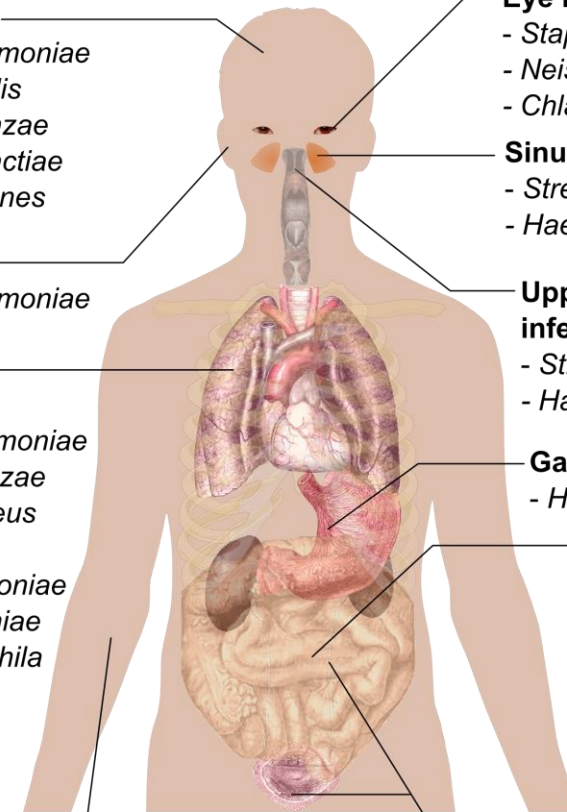
- *Helicobacter pylori*

Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
- *Staphylococcus saprophyticus*
- *Pseudomonas aeruginosa*



Site of Infection

1. ติดเชื้อจุลชีพชนิดอะไร
แบบที่เรื้อรัง

Atypical bacteria:

Mycoplasma pneumoniae: (walking pneumonia)

Very small bacteria lack of peptidoglycan cell wall,

Ability to mimic host cell surface molecules,

Overview of Bacterial infections

Bacterial meningitis

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Otitis media

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Pneumonia

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Atypical:

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- Tuberculosis
- *Mycobacterium tuberculosis*

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Eye infections

- *Staphylococcus aureus*
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Sinusitis

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Upper respiratory tract infection

- *Streptococcus pyogenes*
- *Haemophilus influenzae*

Gastritis

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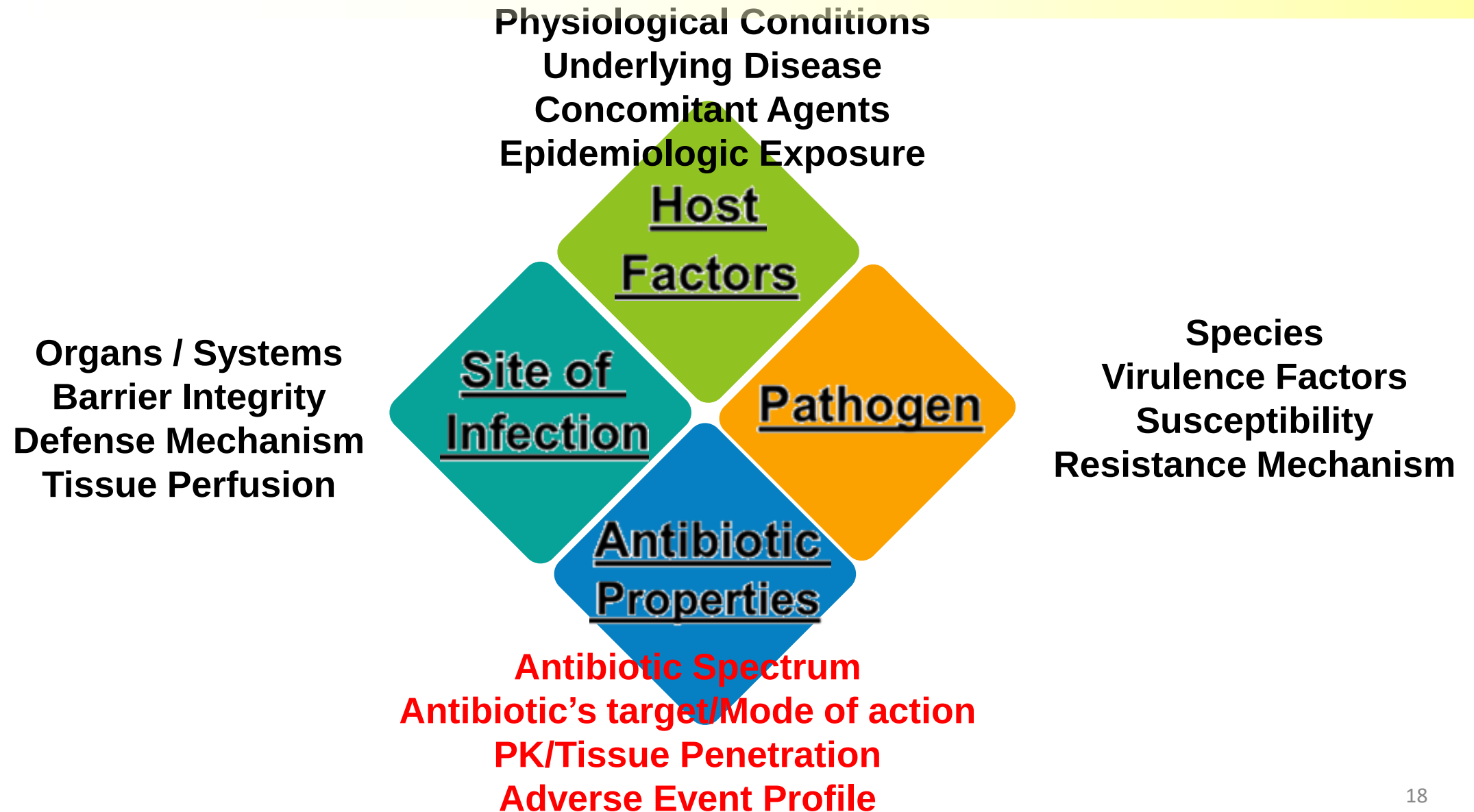
Food poisoning

- *Campylobacter jejuni*
- *Salmonella*
- *Shigella*
- *Clostridium*
- *Staphylococcus aureus*
- *Escherichia coli*

Urinary tract infections

- *Escherichia coli*
- Other Enterobacteriaceae
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Principle of Antimicrobial Therapy



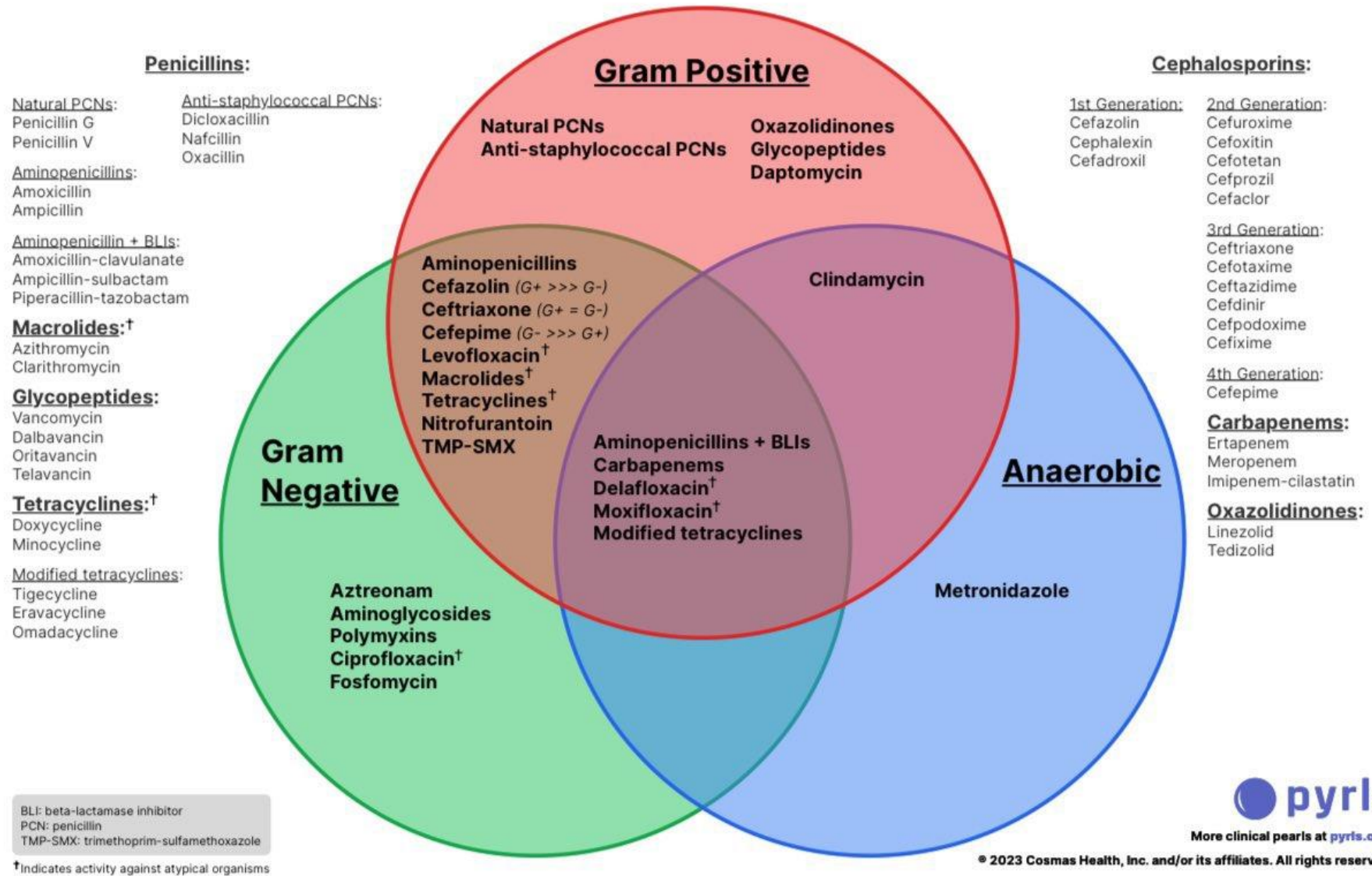
Spectrum of Antimicrobial Drug

	Gram positive cocci			Gram negative bacilli					Gram-negative cocci		Anaerobes	Atypicals
	MRSA	MSSA	Streptococci	E. coli	P. mirabilis	Klebsiella	Pseudomonas	ESCAPPM	N. gonorrhoeae	N. meningitis		
Penicillin			Penicillin G									
Anti-staphylococcal penicillins			Nafcillin/Oxacillin									
Aminopenicillins			Ampicillin/Amoxicillin						Amp/Amox			
1st-gen cephalosporin			Cefazolin, cephalexin									
2nd-gen cephalosporin			Cephotetan, Cefoxitin								Cephotetan, Cefoxitin	
3rd-gen cephalosporin			Ceftriaxone						Ceftriaxone			
			Ceftazidime									
4th-gen cephalosporin			Cefepime									
Aminopenicillins with beta-lactamase inhibitors			Amoxicillin + clavulanate (Augmentin)								Amox-clav	
			Ampicillin + sulbactam (Unasyn)								Amp-sul	
			Piperacillin + tazobactam (Zosyn)						Piperacillin + tazobactam (Zosyn)			
Monobactams			Ertapenem						Ertapenem			
			Imipenem, Meropenem									
Quinolones			Ciprofloxacin						Ciprofloxacin			
			Levofloxacin									Levofloxacin
			Moxifloxacin						Moxifloxacin			
Aminoglycosides			Gent/Tobra/Amikacin									
Lincosamide			Clindamycin								Clindamycin	
Macrolides			Azithromycin						Azithromycin			Azithromycin
Tetracyclines			Doxycycline						Doxycycline			Doxycycline
Glycopeptides			Vancomycin									
Antimetabolite			TMP/SMX (Bactrim)						TMP/SMX			
Nitroimidazoles											Metronidazole	

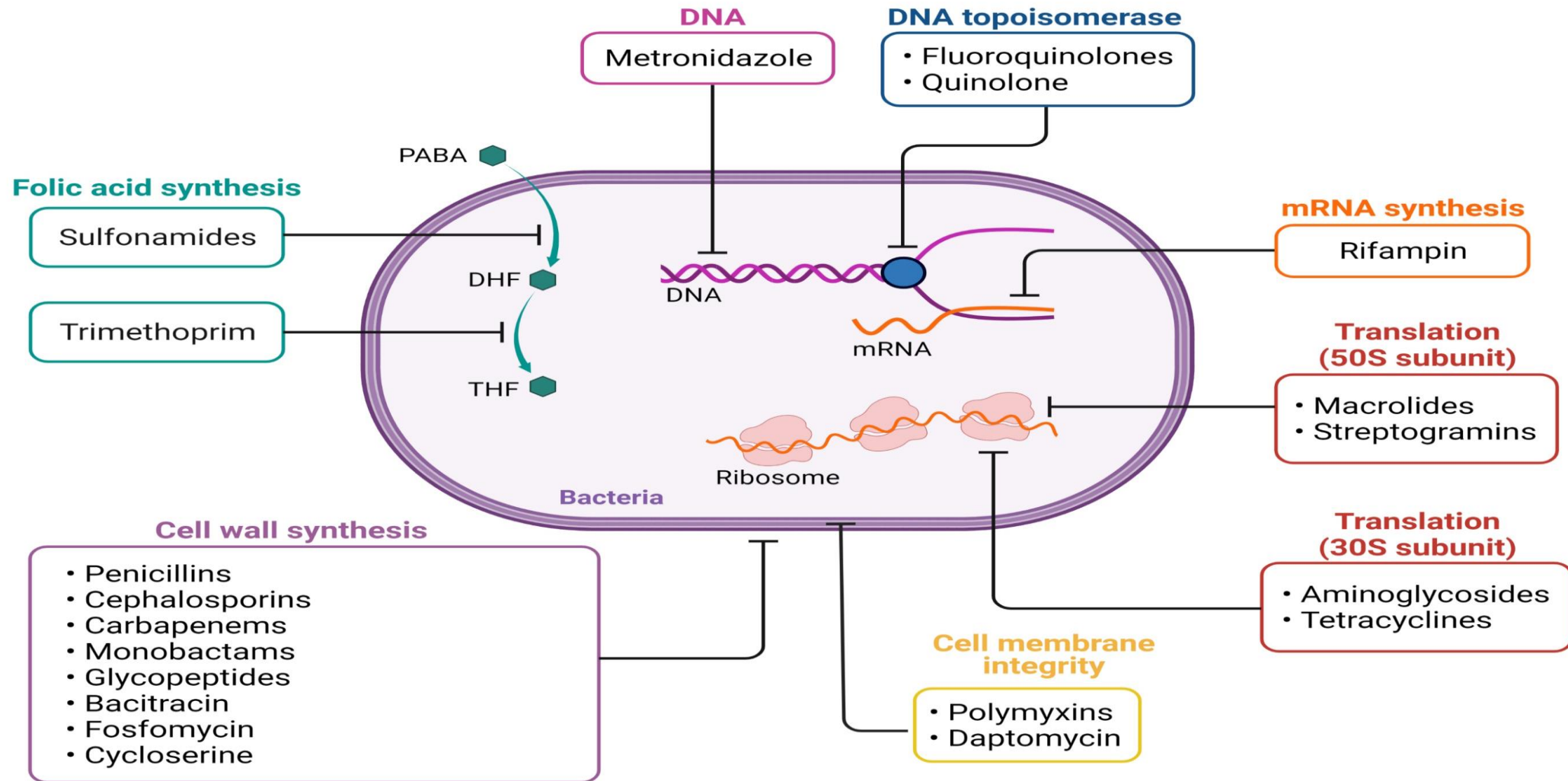
See github.com/aetherist/antibiogram for details. For educational purposes only. TMP/SMX = Trimethoprim-sulfamethoxazole, MRSA = Methicillin-resistant *Staphylococcus aureus*, MSSA = Methicillin-sensitive *Staphylococcus aureus*, ESCAPPM = *Enterobacter* spp., *Serratia* spp., *Citrobacter freundii*, *Aeromonas* spp., *Proteus* spp., *Providencia* spp. and *Morganella morganii*.

Antibiotic Spectrum of Activity: Venn Diagram

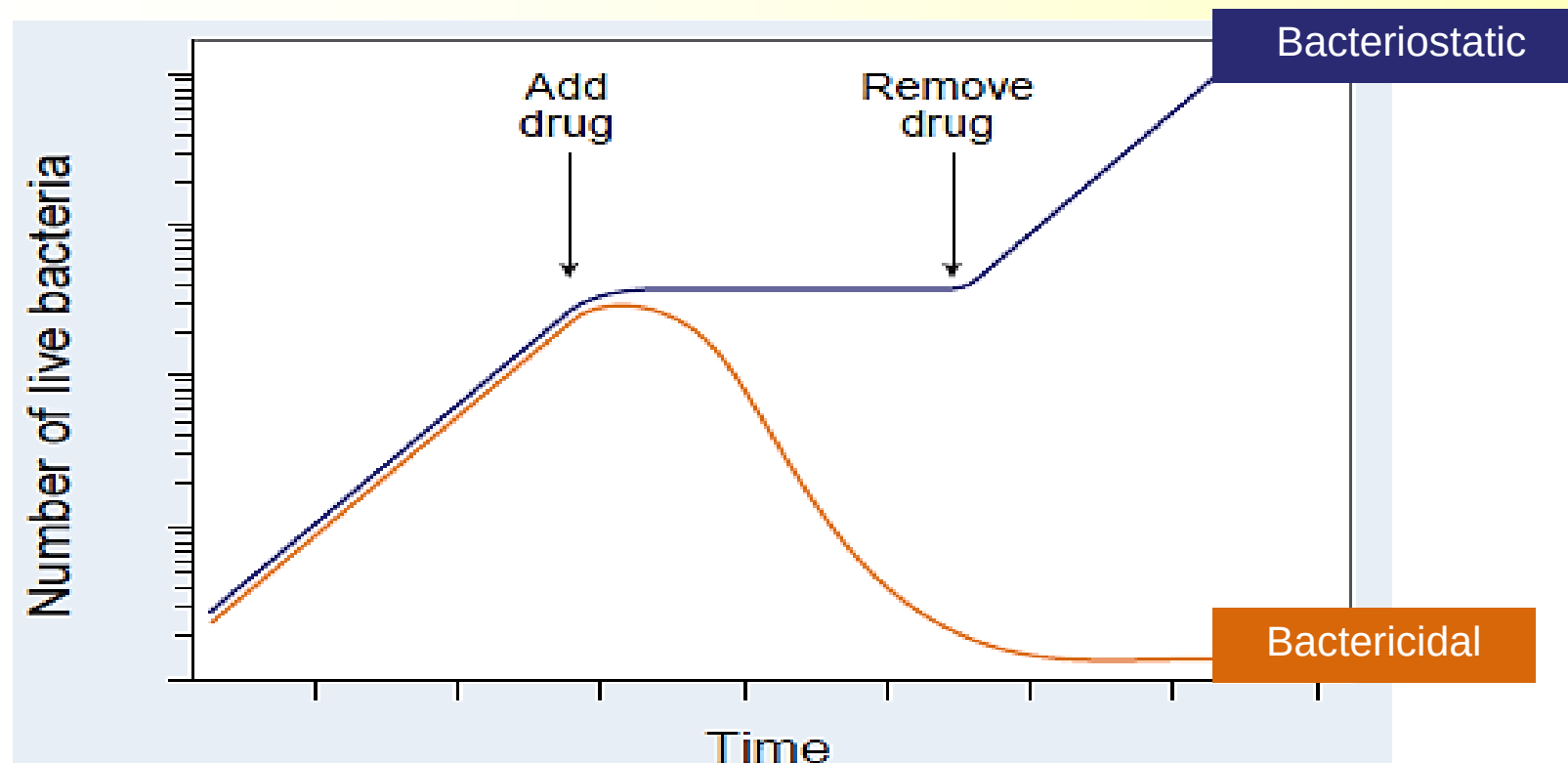
This visual representation provides a concise overview of the general coverage of various classes of antibiotics.
Please note that while this diagram offers valuable insights, it is not exhaustive and should be used as a quick reference rather than a comprehensive resource.



Targets of action of Antimicrobial Drug



(Mode of action),Bacteriostatic vs. Bactericidal



- ◆ Immunocompetent hosts → **Bacteriostatic** and **Bactericidal** drugs are **equivalent** for the treatment of most infectious diseases
- ◆ Immunocompromised, Serious infection, Poor blood flow → **Bactericidal** drugs should be selected to achieve rapid cure

(Mode of action), Bacteriostatic vs. Bactericidal

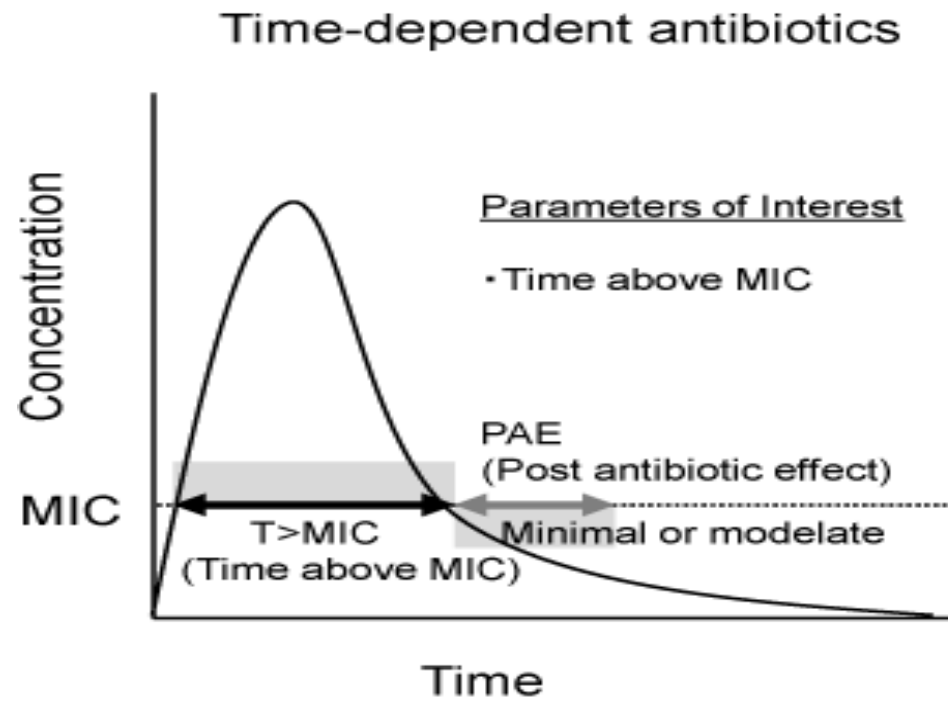
Bactericidal

β -lactam
Metronidazole
Rifampicin
Aminoglycoside
Quinolone
Polymyxin
Trimethoprim
Glycopeptide, Lipopeptide
Linezolid
Tigecycline

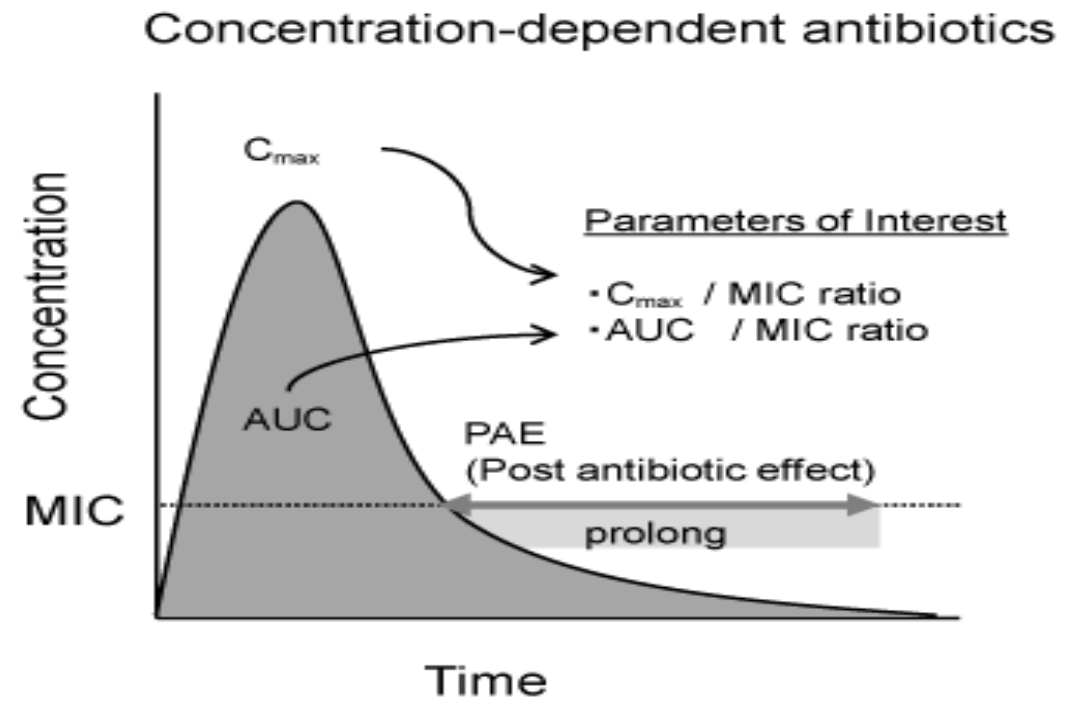
Bacteriostatic

Macrolide
Tetracycline
Lincosamide
Fusidic acid
Chloramphenicol
Oxazolidinone/linezolid

(Mode of action), Time- & Concentration-dependent

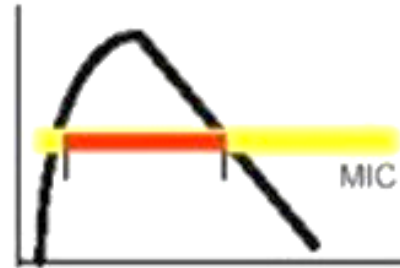
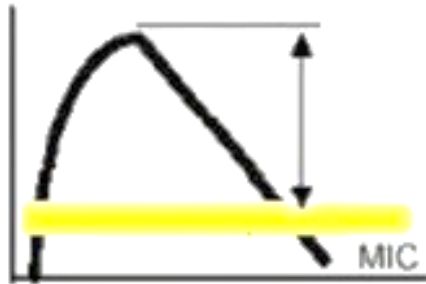


- Killing activity continues as long as serum concentrations are **greater than MIC**



- Killing activity increase with **increasing drug concentrations**
- Responsible for the efficacy of once-daily dosing regimen

Time- & Concentration-dependent



Concentration-dependent	Time-dependent	Concentration- & Time-dependent
♦ Aminoglycosides ♦ Fluoroquinolones ♦ Metronidazole ♦ Polymyxin ♦ Daptomycin	♦ Penicillins ♦ Cephalosporins ♦ Monobactam ♦ Macrolides ♦ Clindamycin	♦ Azithromycin ♦ Vancomycin
Maximize C_{max}/MIC AUC/MIC	Maximize $T > MIC$	Maximize AUC_{0-24}/MIC

Post-antibiotic effect (PAE)

- **Post-antibiotic effect (PAE):** Persistent suppression of bacterial growth that occurs after levels of antibiotic have **fallen below MIC**
- **PAE** reflects time for bacteria to **return to logarithmic growth**

$$\text{PAE} = T - C$$

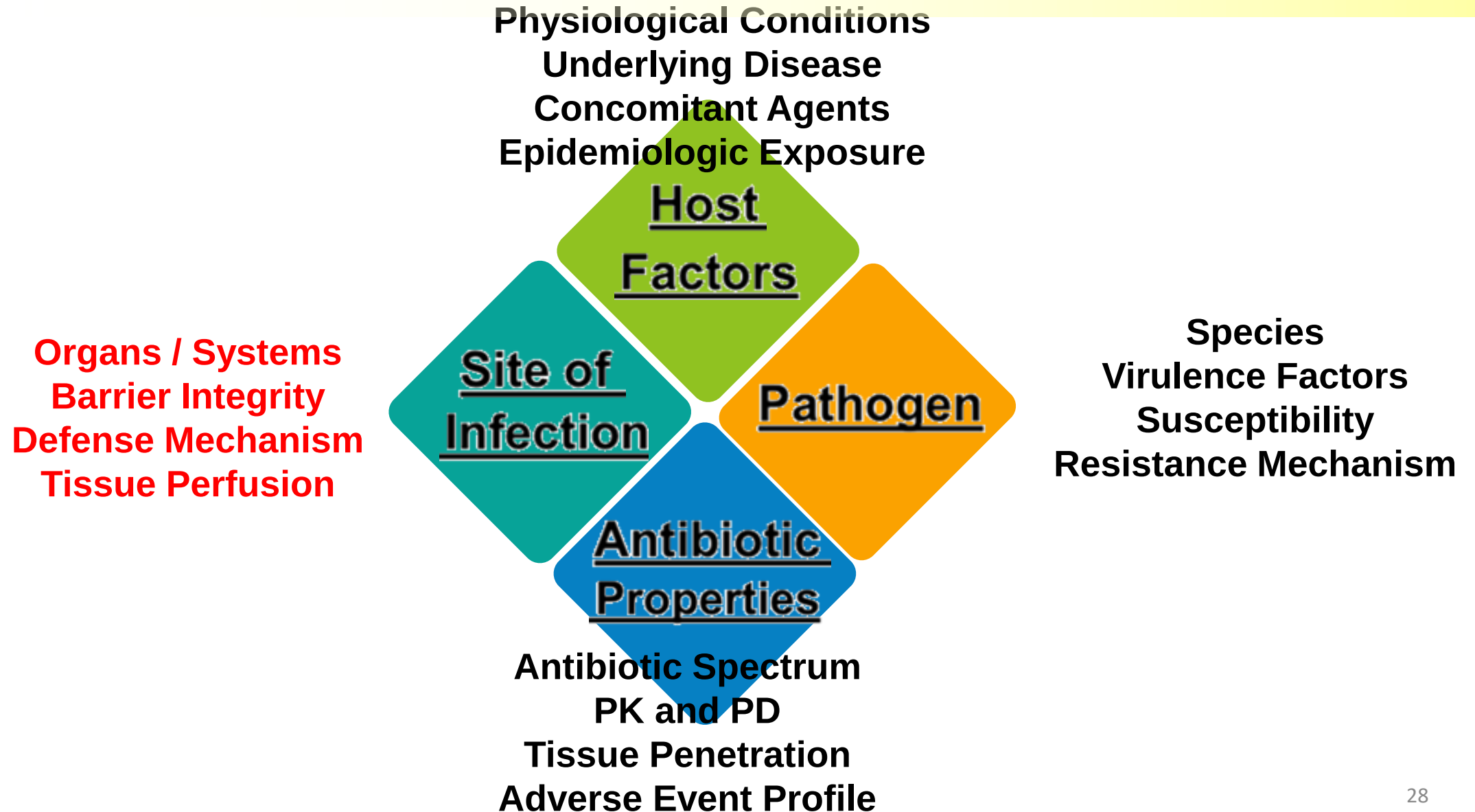
T = time required for bacteria to increase 1 Log₁₀ **after drug removal**

C = time required for bacteria to increase 1 Log₁₀ **without drug treatment**

- **Proposed mechanisms:**
 - Slow recovery after reversible **non-lethal damage** to cell structures
 - **Persistence of drug at a binding site** or within periplasmic space
 - Need to **synthesize new enzymes** before growth can resume

Antibiotics	Gram Positive bacteria	Gram Negative bacteria	<i>Pseudomonas aeruginosa</i>
Penicillins	1-2	0	0
Cephalosporins	1-2	0	0
Carbapenems	1-2	(1)	1-2
Quinolones	1-3	1-3	1-2
Protein synthesis inhibitors	3-5	3-8	
Aminoglycosides		2-4	2-3

Principle of Antimicrobial Therapy



Site of infection

- **Route of Administration**
 - **Oral** vs. **IV**
 - » Bioavailability, Severity, Site of infection, Function of GI tract
- **Tissue Penetrations**
 - **Fluoroquinolone** → high concentration in prostate (except **Moxifloxacin**)
- **Ability to across Blood-brain barrier**
 - » Lipophilicity, Low MW, Low protein binding, Low degree of ionization
 - **1st & 2nd gen Cephalosporins, Macrolides, Aminoglycosides** → **Limited**
- **Special barrier**
 - CSF, ocular fluid, abscess cavity
 - **Aminoglycosides** → inactivated by low pH and low oxygen environment of abscesses

Antimicrobial Therapy

- **Empiric therapy** → Infecting pathogen not yet identified
 - Early intervention to improve outcomes
 - More “**Broad spectrum**” based on experience with clinical trials
- **Definitive therapy** → Infecting pathogen already identified
 - Specific therapy chosen
 - “**Narrow spectrum**” with an **appropriate duration**
- **Prophylactic therapy**
 - Prevention an initial infection or its recurrence after cure
 - Predisposing factors: surgery/procedures, patient conditions

Spectrum of Antimicrobial Drug

	Gram positive cocci			Gram negative bacilli					Gram-negative cocci		Anaerobes	Atypicals
	MRSA	MSSA	Streptococci	E. coli	P. mirabilis	Klebsiella	Pseudomonas	ESCAPPM	N. gonorrhoeae	N. meningitis		
Penicillin			Penicillin G									
Anti-staphylococcal penicillins			Nafcillin/Oxacillin									
Aminopenicillins			Ampicillin/Amoxicillin						Amp/Amox			
1st-gen cephalosporin			Cefazolin, cephalexin									
2nd-gen cephalosporin			Cephotetan, Cefoxitin								Cephotetan, Cefoxitin	
3rd-gen cephalosporin			Ceftriaxone						Ceftriaxone			
			Ceftazidime									
4th-gen cephalosporin			Cefepime									
Aminopenicillins with beta-lactamase inhibitors			Amoxicillin + clavulanate (Augmentin)								Amox-clav	
			Ampicillin + sulbactam (Unasyn)								Amp-sul	
			Piperacillin + tazobactam (Zosyn)								Piperacillin + tazobactam (Zosyn)	
Monobactams			Ertapenem								Ertapenem	
			Imipenem, Meropenem									
Quinolones			Ciprofloxacin									
			Levofloxacin									Levofloxacin
			Moxifloxacin								Moxifloxacin	
Aminoglycosides			Gent/Tobra/Amikacin									
Lincosamide			Clindamycin								Clindamycin	
Macrolides			Azithromycin								Azithromycin	
Tetracyclines			Doxycycline								Doxycycline	
Glycopeptides			Vancomycin									
Antimetabolite			TMP/SMX (Bactrim)									
Nitroimidazoles											Metronidazole	

See github.com/aetherist/antibiogram for details. For educational purposes only. TMP/SMX = Trimethoprim-sulfamethoxazole, MRSA = Methicillin-resistant *Staphylococcus aureus*, MSSA = Methicillin-sensitive *Staphylococcus aureus*, ESCAPPM = *Enterobacter* spp., *Serratia* spp., *Citrobacter freundii*, *Aeromonas* spp., *Proteus* spp., *Providencia* spp. and *Morganella morganii*.

Empiric therapy

คู่มือการใช้ยาต้านจุลชีพ

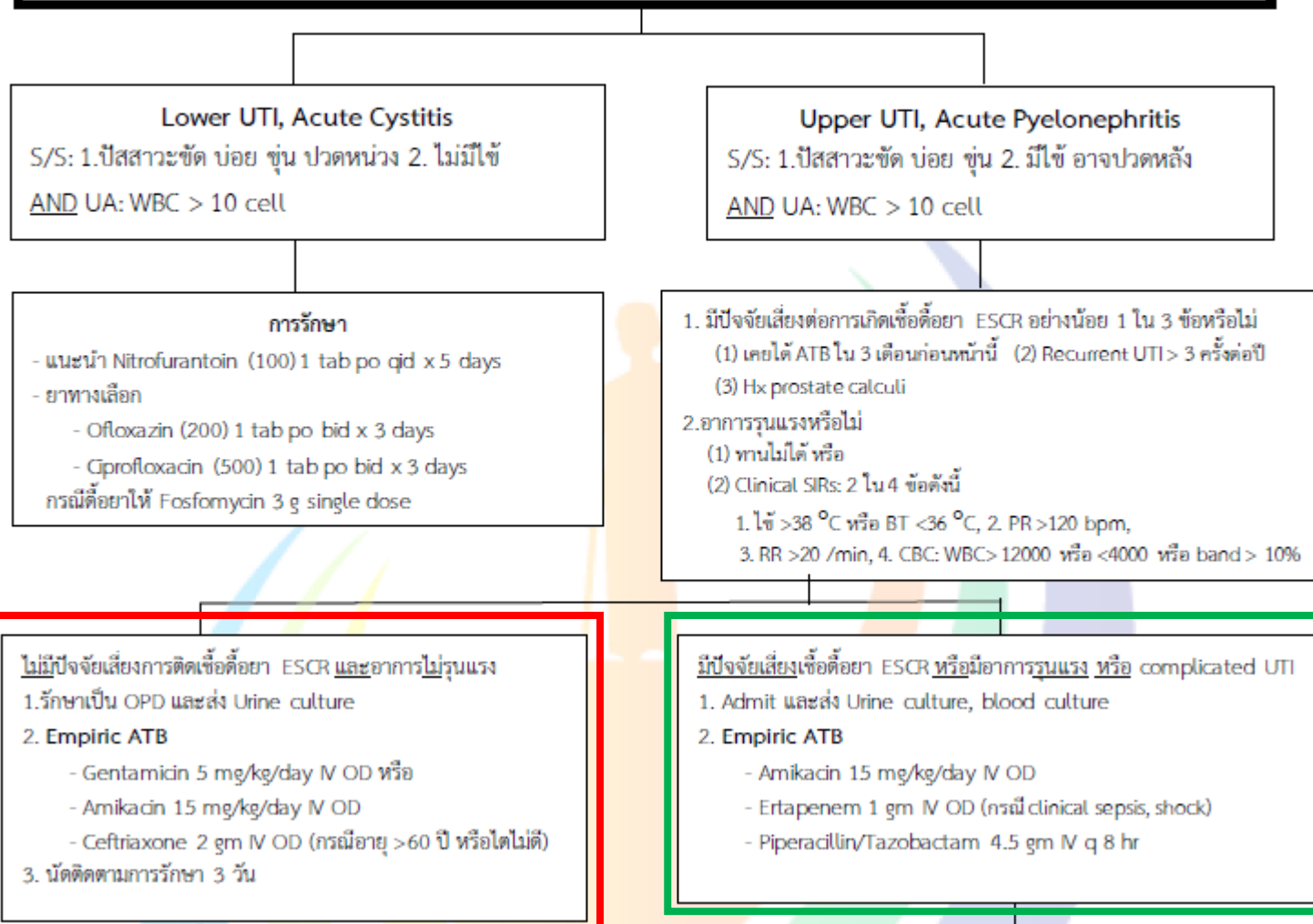
อย่างสมเหตุผล

โครงการสนับสนุนและส่งเสริมการใช้ยาต้านจุลชีพ
อย่างสมเหตุผล (Antimicrobial Stewardship Program)

โรงพยาบาลมหาวิทยาลัยราชภัฏวชิร



แนวทางการรักษาโรคติดเชื้อทางเดินปัสสาวะในผู้ใหญ่ (Community acquired-UTI) โรงพยาบาลมหาวิทยาลัยราชภัฏวชิร

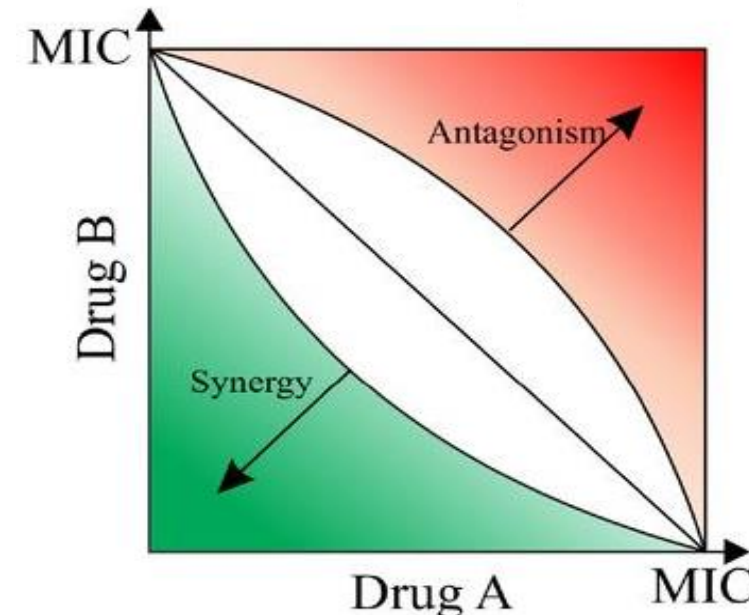


Antimicrobial Therapy

- **Monotherapy**
 - reduce risk for drug interaction, medication error, missed dose
- **Combination therapy**
 - **Polymicrobial infections**
 - Cover **common known/suspected** pathogen, but **not** all possible pathogens e.g. intra-abdominal abscesses (Gr(-)+anaerobic+enterococci)
 - **Decrease dose-related toxicity**
 - Flucytosine + Amphotericin B → ↓ Amphotericin B-induced nephrotoxicity
 - **Decrease drug resistance** (e.g. Tuberculosis)
 - **Enhance activity** (synergism)
 - **More broad-spectrum empiric therapy** (life-threatening infection)

Combination therapy

- Synergism:** Combination results in **more than 4-fold reduction** in MIC or MIB compared to each drug alone
 - **Trimethoprim + Sulfamethoxazole** → block 2 step in folic acid pathway
 - **β -lactam + β -lactamase inhibitor** → Inhibit enzymatic inactivation
 - **Penicillin/Ampicillin + Gentamicin/Streptomycin** → increase uptake
- Antagonism:** Killing effects of combined drug are **less than** each drug alone
 - **Cell wall inhibitor + bacteriostatic agent** → Inhibit activity of cell wall-active agents which require bacteria to be active growing



การเลือกใช้ยาต้านจุลชีพ (กรณีพิเศษ)

1. ติดเชื้อจากสถานที่ใด: ทะเล

ใช้เป็นข้อมูลบ่งบอกชนิดของเชื้อที่น่าจะเป็นสาเหตุ

Relative contribution (%) of taxonomic groups to the culturable fraction of the sea surface microlayer (SML) and underlying waters (UW) bacterial communities from Banyuls-sur-Mer and Barcelona

	Barcelona		Banyuls-sur-Mer	
	SML	UW	SML	UW
<i>Alphaproteobacteria</i>	19	18	19	24
<i>Gammaproteobacteria</i>	24	35	53	47
CFB group	8	12	13	16
Total Gram-negative	51	65	85	87
Actinobacteria	27	6	6	3
Firmicutes	22	29	9	10
Total Gram-positive	49	35	15	13

Hans-Curt Flemming and Stefan Wuerzt. Nat Rev MicroBiol 2019; 17: 247-60

- ***Candidatus Pelagibacter ubique***: Gram-, abundance in summer
- ***Vibrio vulnificus***: Gram-, flagella, capsule, exotoxin (MARTX-multifunction autoprocessing repeats-in-toxin), infection causes rapidly cellulitis and sepsis
- ***Synechococcus spp.* Gram-**



<https://www.aafp.org/afp/2007/0815/afp20070815p539.pdf>



Linhui li., et al. IDcase 2018: e00497

Vibrio vulnificus infection

Symptoms	Treatment
Gastroenteritis	Ciprofloxacin or Norfloxacin or Teracyclin
Wound infection	(Ceftazidime or Cefotaxime) + (Doxycycline or Ciprofloxacin)
Septicemia	(Ceftazidime or Cefotaxime) + (Doxycycline or Ciprofloxacin)+ ICU supportive care

Antibiotic stability and solubility

Generic name (Brand name)	Route	Reconstitution		Stability after Reconstituion/Dilution		Diluent		หมายเหตุ
		Solvent (ตัวทำละลาย)	Final conc. (ความเข้มข้นหลังละลายยา)	Room temp.	2-8° C	Solvent (สารละลายสำหรับเจือจางยา)	Final conc. (ความเข้มข้นหลังเจือจาง)	
Liposomal Amphotericin-B (Ambisome®)	INF	SWFI	50 mg/ 12 ml	Protect from light 24 hrs (dilute) 14 days (recon & dilute)		only D5W	1-2 mg/ml (adult) 0.2-0.5 mg/ml (child)	ห้ามใช้ Inline filter เพราะยาจะแตกออกจาก liposome
Ampicillin sodium (Ampin Momac®)	IV, INF, IM	SWFI	IM 250 mg/ml 1 g/ 3.5 ml IV 250 mg/ 5 ml 1 g/ 10 ml	Immediately use		only NSS ห้ามใช้ D5W (degrade)	≥ 3 g/ 150 ml	ห้ามใช้ D5W (degrade)
Ampicillin + Salbactam (Unasyn®)	IV, INF, IM	SWFI	3 g/ 6.4 ml	มีความคงตัว 1 hrs (หลัง reconstitution) กรณี IM,IV ควรรีบใช้ทันที ส่วนกรณี INF ควรรีบ Dilution 8 hrs (dilute)	NA	D5W, NSS, D5S/2	30-45 mg/ml (Unasyn® Dose/ml) (50-100 ml of sol)	

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อาการไม่พึงประสงค์ของการใช้ยาต้านจุลชีพ

Hypersensitivity:

Urticaria, anaphylaxis, TEN, DRESS, maculopapular exanthema, angioedema

Direct toxic:

Nephrotoxic, hepatotoxic

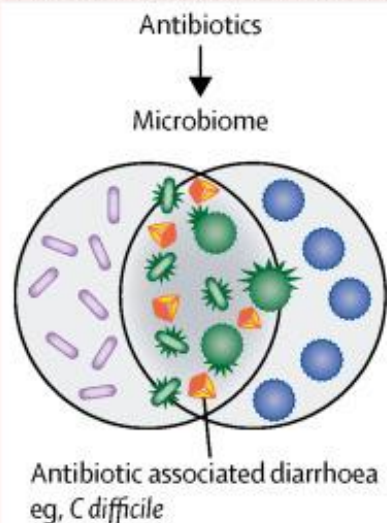
Superinfection:

pseudomembranous colitis จากการใช้ broad spectrum antibiotics

On-target ADRs

Predictable based on drug action*

ADR mechanisms



Off-target ADRs

Non-immunologically-mediated HSRs

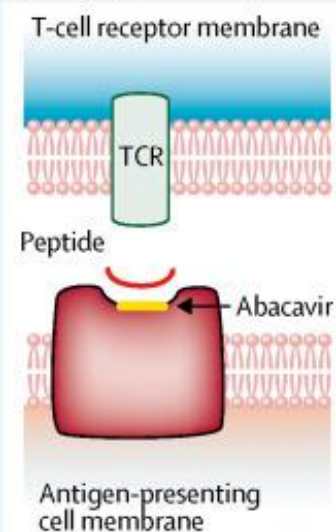
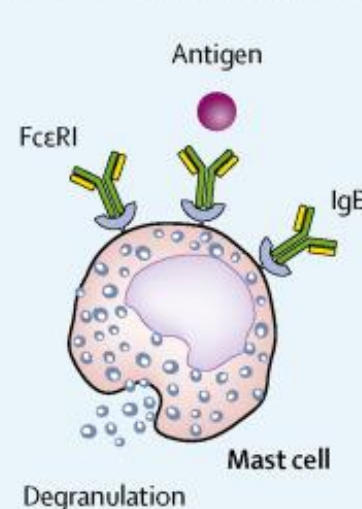
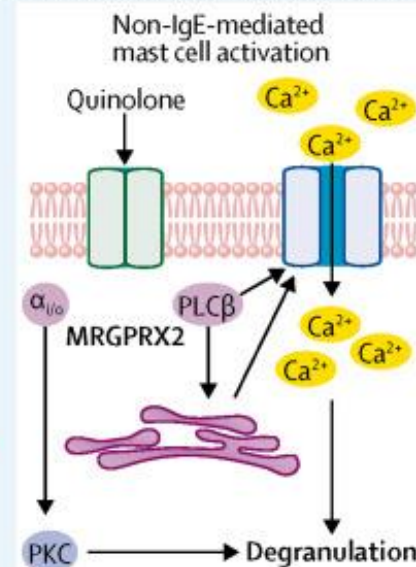
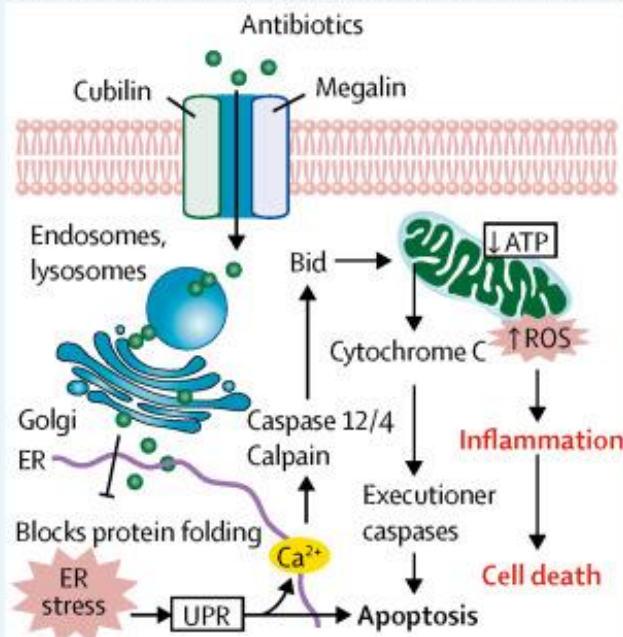
Cellular toxicity and disrupted physiology*
Non-immune cell receptor interaction*

Immune receptor interaction*

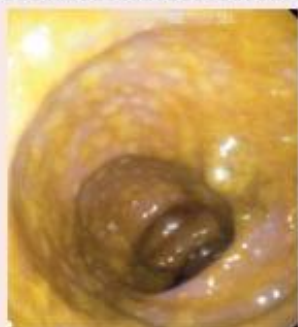
Immunologically-mediated HSRs

Antibody-mediated

Pure T-cell-mediated*



ADR phenotype/example



C difficile associated pseudomembranous colitis



Aminoglycosides acute tubular necrosis



Fluoroquinolone urticaria

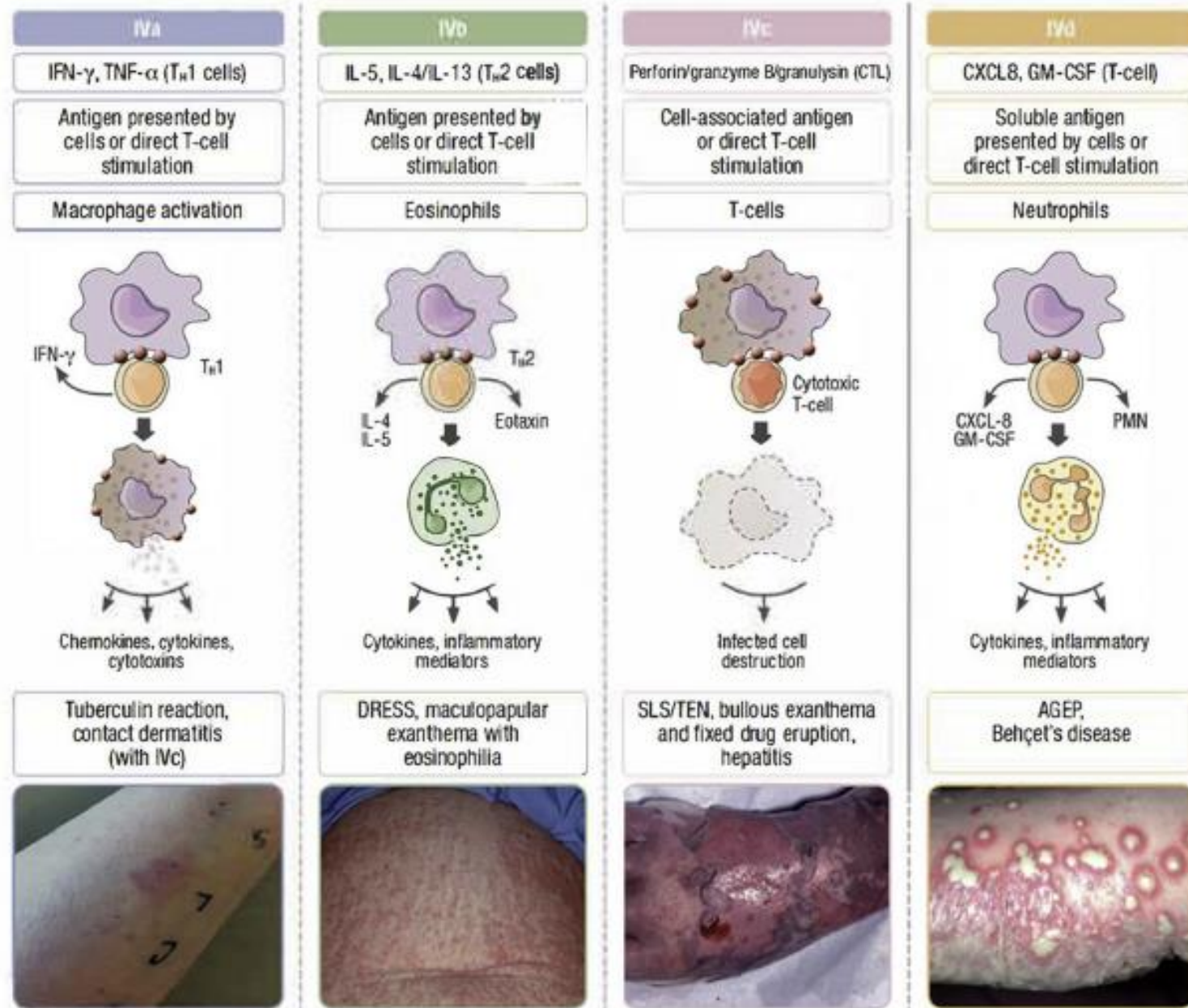


Penicillin angioedema

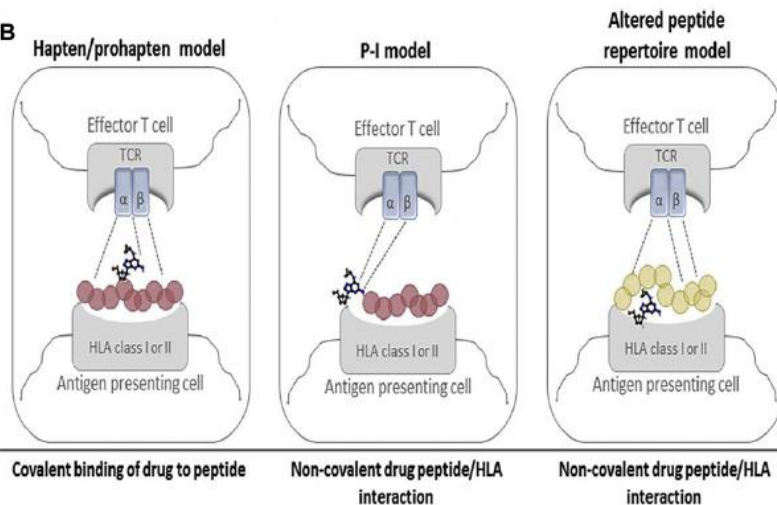


Abacavir hypersensitivity

A



B

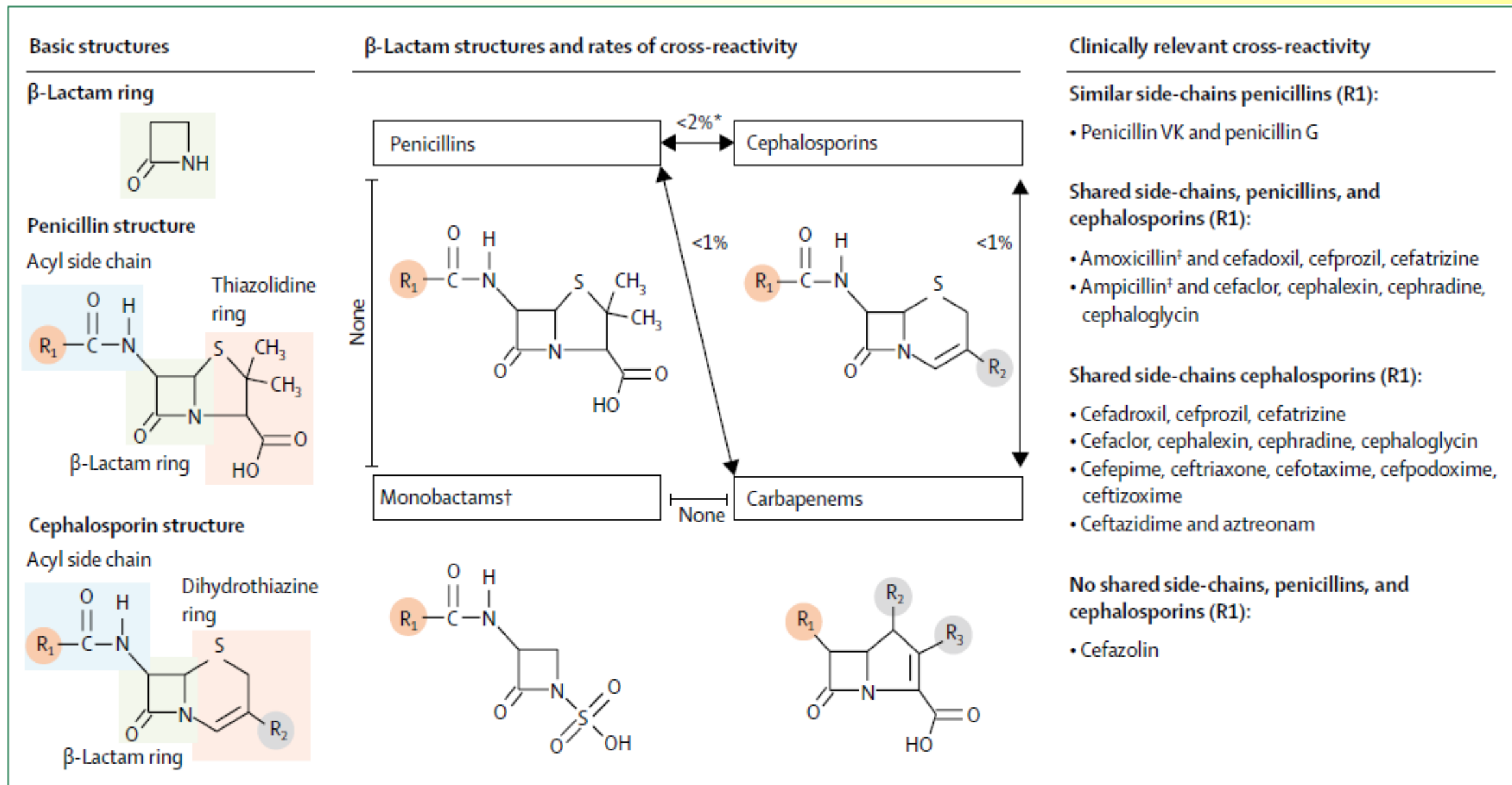


Neo-epitope formed by drug binding to peptide

Drug bind to peptide/HLA at cell surface

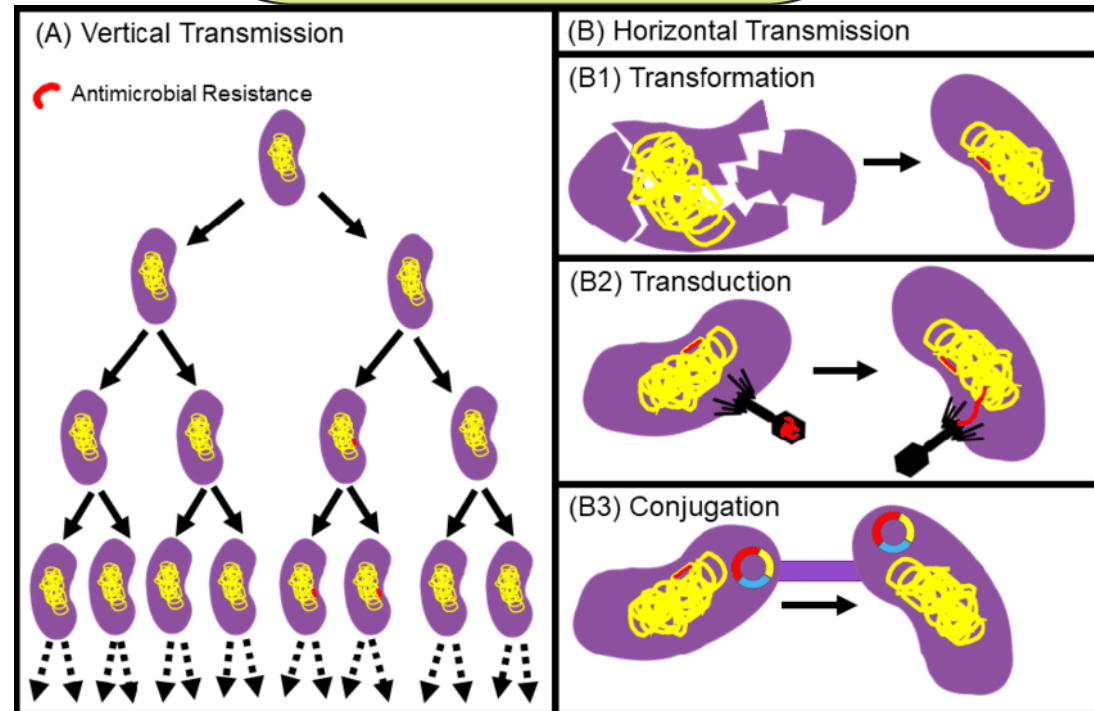
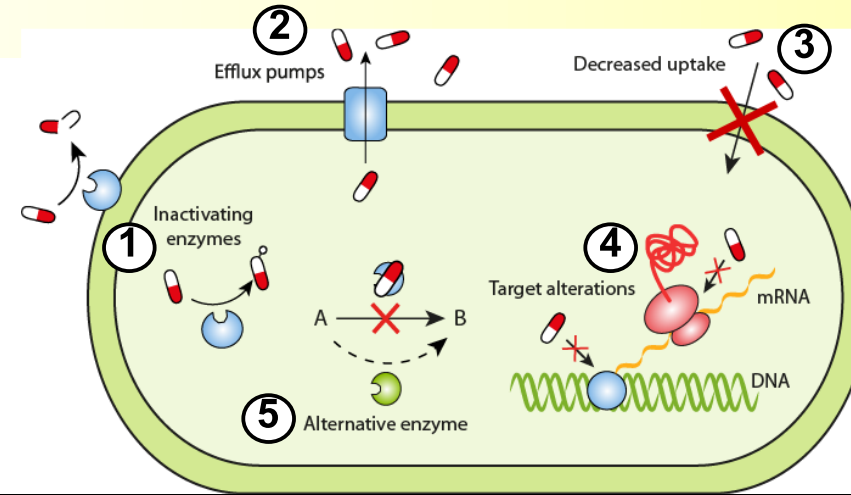
Drug binding results in a change in HLA binding motif and a selection of altered peptides

อาการไม่พึงประสงค์ของการใช้ยาต้านจุลชีพ



Antimicrobial Resistance

- **Natural (Inherent)**
 - Gr(-) LPS
 - Lack target or transport
- **Acquired**
 - Mutation
 - Horizontal transfer
- **Vertical**
- **Horizontal**
 - Transformation
 - Transduction
 - Conjugation



Mechanism of Resistance

	β-LACTAM	AMINO-GLYCOSIDE	CHLORAM-PHENICOL	MACROLIDE	SULFONAMIDE	TETRACYCLINE
Enzymatic inactivation	+++	+++	+++	+ (gram-negative)	–	–
Decreased permeability	+ (gram-negative)	+ (gram-negative)	+ (gram-negative)	++ (gram-negative)	–	+ (gram-negative)
Efflux	+	+	+	++	–	+++
Alteration of target site	++	++	–	+++	++	+ (<i>Helicobacter pylori</i>)
Protection of target site		–	–	–	–	++
Overproduce target	–	–	–	–	++	–
Bypass of inhibited process	–	–	–	–	+	–
Bind up antibiotic	–	–	–	–	–	–

+++ , most common mechanism; ++, common; + less common.

Mechanism of Resistance

	TRIMETHOPRIM	QUINOLONE	GLYCOPEPTIDE	LINOSAMIDE; STREPTOGRAMIN	RIFAMPIN
Enzymatic inactivation	-	+	-	-	-
Decreased permeability	+ (gram-negative)	+ (gram-negative)	++ (gram-negative)	+ (gram-negative)	-
Efflux	-	+	-	-	-
Alteration of target site	+++	+++	+++	+++	+++
Protection of target site	-	+	-	-	-
Overproduce target	++	-	+	-	-
Bypass of inhibited process	+	-	-	-	-
Bind up antibiotic	-	-	++	-	-

Mechanism of Resistance

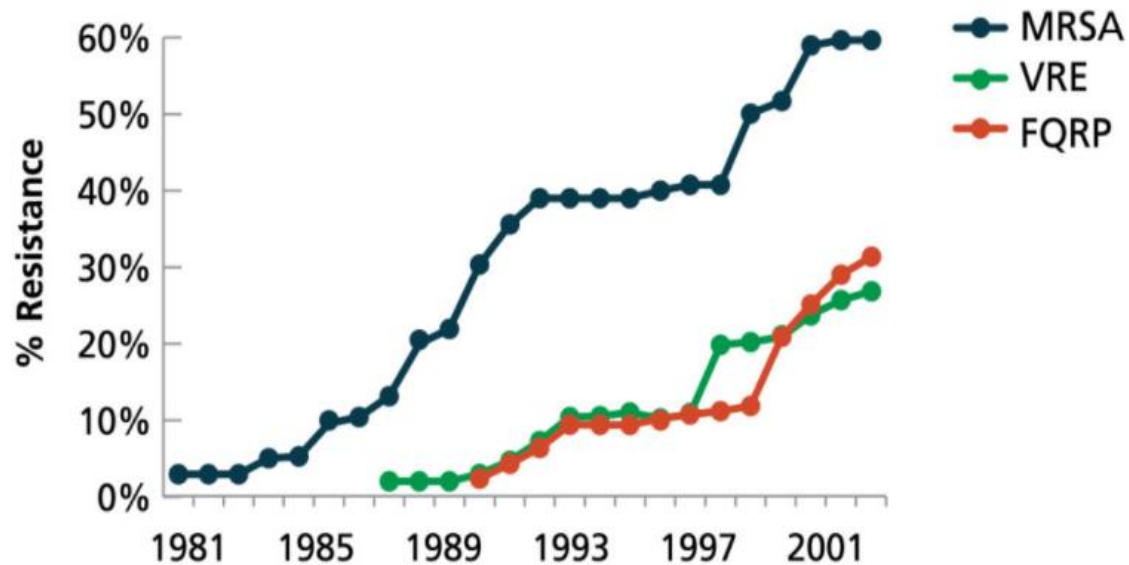
	POLYMYXIN	DAPTOMYCIN	LINEZOLID
Enzymatic inactivation	–	–	–
Decreased permeability	–	+ (gram-negative)	+ (gram-negative)
Efflux	+	–	+
Alteration of target site	++	++	++
Protection of target site	–	+	–
Overproduce target	–	–	–
Bypass of inhibited process	–	–	–
Bind up antibiotic	++	–	–

+++ , most common mechanism; ++, common; + less common.

Mechanism of Resistance

	QUINUPRISTIN- DALFOPRISTIN	METRONIDAZOLE	TIGECYCLINE
Enzymatic inactivation	++	–	–
Decreased permeability	–	–	+
Efflux	+	–	++
Alteration of target site	+	–	–
Protection of target site	–	–	–
Overproduce target	–	–	–
Bypass of inhibited process	–	++	–
Bind up antibiotic	–	–	–

INCREASE IN ANTIBIOTIC RESISTANCE (NNIS 1999)



MRSA-Methicillin Resistant Staphylococcus Aureus

VRE-Vancomycin-Resistant Enterococci

FQRP-Fluoroquinolone-Resistant Pseudomonas aeruginosa

ESBL (Extended Spectrum Beta-Lactamases)

- Hospital acquired
- Carbapenem- are reserved for ESBL infection
- Bacteria
 - E. coli
 - K. pneumoniae
 - P. aeruginosa
 - A. baumannii
 - S. typhimurium
 - Enterobacteriaceae



ดาบ
เพนนิซิลลิน
ต่อสู้
ศัตรูร้าย
กรัมบวก

ดาบ
เซฟาโลสปอริน
ต่อสู้
ศัตรูร้าย
กรัมบวก
กรัมลบ

ดาบ
คาร์บาพีแนม
ต่อสู้
ศัตรูร้าย
ดื้อยา
ESBL

ดาบโคลิสติน
ดาบสุดท้าย
ของมนุษยชาติ
ที่ใช้ต่อสู้
บอสของศัตรู
A.baum ได้

Spartan Doctor



คนเอามาใช้
เลี้ยงฟาร์มหมู
ชะฉิบ



ใช้ยาสมเหตุผล
ไม่จน ไม่แพ้ ไม่ดื้อยา